

PrintTac: Reproducible Tactile Fingertips for Scalable Robot Learning

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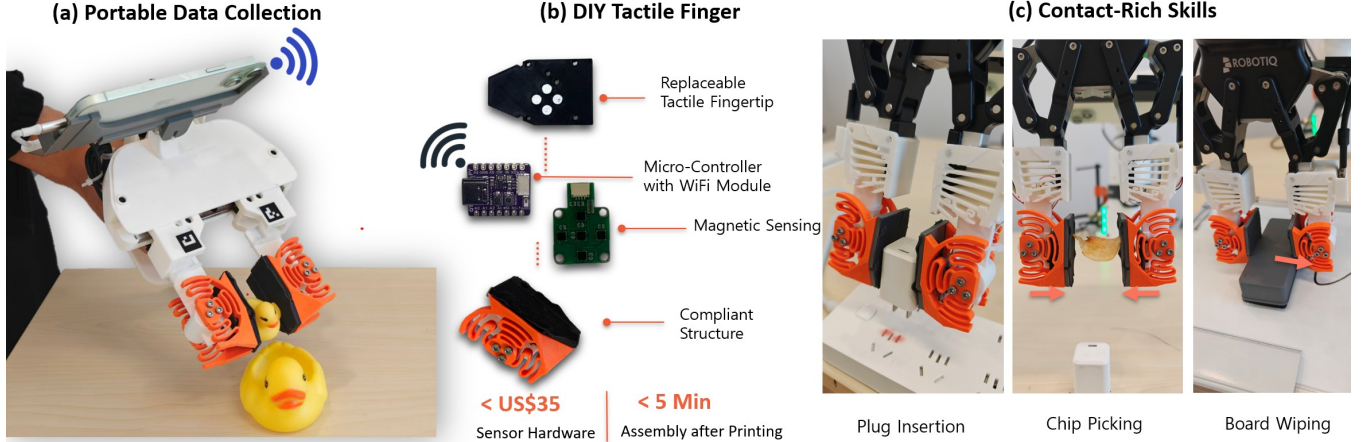


Fig. 1: **Overview of the proposed system.** (a) Portable data collection using an iPhone-equipped UMI with PrintTac fingers. (b) Modular tactile-finger design with a replaceable fingertip, magnetic sensing electronics, a microcontroller, and a 3D-printed compliant structure. (c) Contact-rich manipulation tasks: plug insertion, chip picking, and whiteboard wiping.

Abstract—While tactile feedback is critical for contact-rich manipulation, scaling tactile robot learning remains bottlenecked by hardware costs, fabrication complexity, and sensor-to-sensor domain gaps that hinder data reuse. High-resolution tactile sensors incur substantial visual-encoding overheads, while existing low-cost magnetic alternatives rely on elastomer casting with limited manufacturing consistency. We present PrintTac, a low-cost (<USD 35), DIY-reproducible tactile fingertip designed for consistent, scalable multimodal data collection. By coupling an off-the-shelf magnetometer board with a monolithic 3D-printed compliant mechanism, PrintTac replaces cast elastomers with a reproducible flexure, eliminating elastomer mixing, casting, and curing. It can be assembled in five minutes without per-device recalibration and features a modular, quickly replaceable contact tip. Permanent magnets on the rigid contact plate undergo shared 6-DoF rigid-body motion, captured as magnetic-field measurements by three-axis magnetometers. Force-estimation benchmarks show that models trained on a single sensor transfer directly to independently fabricated instances with minimal error. We further integrate PrintTac into an untethered UMI data-collection system that synchronizes tactile signals, visual streams, and ARKit poses onboard a smartphone over Wi-Fi, eliminating the need for a host PC. Real-robot experiments on insertion, grasping, and surface-interaction tasks show that PrintTac-conditioned policies consistently outperform vision-only baselines trained on the same demonstrations.

I. INTRODUCTION

Imitation learning has enabled robots to acquire diverse manipulation behaviors directly from physical demonstrations [1], [2]. However, visual observations can leave contact states ambiguous when the interaction region is occluded. Visuo-tactile manipulation studies show the value of local contact information in such settings [3], while force-aware

policies have been evaluated on tasks requiring careful force modulation, including wiping and insertion [4]. Making this feedback practical for robot learning also requires attention to sensor integration, replacement, and data reuse [5].

Fabrication, sensing performance, and cross-instance consistency are important considerations in tactile sensor design. Vision-based tactile sensors such as GelSight and DIGIT provide detailed contact observations using cameras, illumination, and compliant optical interfaces [6], [7]. Magnetic tactile sensors offer another approach: ReSkin uses a magnetized elastomer skin [8], while other designs track permanent magnets embedded in a deformable medium [9]. Recent work has improved both fabrication and replacement: AnySkin demonstrates zero-shot transfer of learned manipulation policies across sensor instances [5], and eFlesh uses 3D-printed microstructures to customize sensor geometry and mechanical response [10]. Building on these advances, we focus on a tactile fingertip whose compliant sensing structure and task-specific contact interface can be designed separately, and which can be integrated into a portable demonstration-collection system.

In this work, we argue that scalable tactile learning necessitates a sensor design governed by mechanical determinism and representational simplicity. We present *PrintTac*, a low-cost, DIY-reproducible tactile fingertip engineered specifically for consistent cross-device performance and high-throughput data collection. Instead of relying on manual elastomer synthesis, PrintTac leverages a monolithic, 3D-printed compliant mechanism paired with an off-the-shelf array of five three-axis magnetometers. External contact

forces deform the compliant flexure, inducing 6-DoF motion at the contact surface that is implicitly captured by the magnetometer array through the resulting magnetic-field variations. Each flexure is fabricated from the same CAD model using a fixed set of printing parameters. Furthermore, the contact interface is mechanically decoupled from the internal sensing core, allowing task-specific contact pads to be swapped in under one minute without disturbing the magnetic circuit or requiring sensor recalibration.

To bring PrintTac from an isolated sensor to a learning pipeline, we develop an untethered, handheld data-collection system building upon the Universal Manipulation Interface (UMI) framework [11]. UMI-FT combines an iPhone with finger-level force/torque sensing, but its published implementation records force/torque data through a USB connection to a laptop [4]. Our system instead routes tactile measurements from a Wi-Fi microcontroller directly to the handheld smartphone. The phone records visual streams and end-effector trajectories from onboard spatial tracking together with the tactile measurements, enabling demonstration collection without a tethered workstation.

Our main contributions are summarized as follows:

- **A Reproducible, Deterministic Tactile Fingertip:** We introduce PrintTac, a low-cost (<USD 35), modular tactile sensor that bypasses elastomer molding in favor of a monolithic 3D-printed flexure. Its deterministic mechanics minimize fabrication tolerances, while its decoupled contact surface permits sub-minute swaps without recalibration.
- **Cross-Instance Consistency without Calibration:** We evaluate cross-instance transfer by training force-estimation models on one PrintTac fingertip and testing them on a separately fabricated fingertip.
- **An Untethered Multimodal Demonstration Suite:** We present a completely wireless, handheld data-collection pipeline that pairs PrintTac with a commercial smartphone, streaming and synchronizing multimodal observations without tethered workstations.
- **Real-Robot Manipulation Validation:** Across contact-intensive tasks including plug insertion, delicate chip grasping, and whiteboard wiping, we show that policies trained with PrintTac data achieve higher task success than vision-only baselines trained on the same demonstrations.

II. RELATED WORK

For tactile sensing to scale from a sensor prototype to a robot-learning system, two requirements must be addressed together. Contact must first be encoded in a form that a policy can use, and that encoding must remain meaningful when demonstrations and deployment involve different sensor instances. Portable demonstration interfaces make this connection especially important because the data-collection device and the robot are physically separate. We review prior work along this progression, from tactile observations, to cross-instance sensing, and finally to multimodal demonstration collection.

A. Tactile Sensing and Representations for Manipulation

Vision-based tactile sensors make local contact observable through images of a deformable surface. GelSight [6] established this approach for high-resolution geometry and force measurement, while GelSlim [12] and DIGIT [7] brought it into more compact robotic fingertips. Later systems increased the information captured at contact: 9DTact [13] recovers surface geometry and six-axis force, while PolyTouch [14] combines tactile images with acoustic and peripheral visual observations. The resulting observations are rich, but they also shift part of the learning problem to representation design. T-Dex [15] learns an image-based tactile encoder from robotic play; 3D-TacDex [16] imposes a canonical representation on distributed tactile measurements; and 3D-ViTact [17] aligns dense visual and tactile observations in a shared 3D space. Together, these works improve what a policy can infer from touch. A representation learned on one sensor, however, does not by itself ensure that the same contact will produce a comparable observation after that sensor is replaced. This moves the question from representation alone to the physical consistency of the sensing interface.

B. Magnetic Tactile Sensing and Cross-Instance Reuse

Magnetic sensing makes this physical consistency explicit: the observation depends on how contact deforms magnetic material relative to the field sensors. ReSkin [8] separates a replaceable magnetized elastomer from the electronics and models variation across samples and time. AnySkin [5] advances the same line toward adhesive-free replacement and, crucially, demonstrates zero-shot policy transfer across independently fabricated skins. These results establish cross-instance reuse as a concrete learning criterion rather than only a fabrication property.

Discrete-magnet sensors expose a complementary design trade-off. The magnetic soft tactile sensor of Mohammadi et al. [9] embeds permanent magnets in a deformable medium to localize contact, while the sensor of Rehan et al. [18] uses this principle for multi-axis force estimation. In both cases, the signal is coupled to the geometry and mechanics of the deformable medium. eFlesh [10] uses 3D-printed cut-cell microstructures to control this relationship, enabling customized geometry and mechanical response while evaluating consistency across printed instances. PrintTac builds on this fabrication direction but separates the task-specific rigid contact tip from the printed compliant mechanism. The distinction matters for learning: changing the contact geometry need not require redesigning the full sensing core, while replacing the sensing instance still presents a direct test of mechanical reproducibility. We therefore evaluate both force-model transfer and fixed-policy deployment on independently fabricated instances, explicitly reporting which instances are used for demonstration, training, and deployment.

C. Portable Multimodal Demonstration Collection

Cross-instance consistency becomes consequential when demonstrations are collected away from the robot. UMI [11] separates data acquisition from robot execution by using a

handheld gripper to record visual observations and gripper trajectories. Visuo-tactile interfaces extend this separation to contact-rich tasks. ViTaMin [19] integrates camera-based tactile fingers and onboard recording into a handheld device, while Touch in the Wild [20] uses a portable gripper to collect synchronized visual and tactile observations and learn cross-modal representations. FreeTacMan [21] moves the interface to a wearable form and recovers motion through optical tracking. These systems show that touch can be collected without operating the target robot, but they also introduce a hardware boundary between the device that records contact and the device that later executes the policy.

Recent UMI variants narrow this boundary by collecting force or tactile signals together with pose. UMI-FT [4] pairs an ARKit-tracked iPhone with finger-level force/torque sensors; in its published implementation, those measurements are logged to a laptop through a USB–serial connection. TacUMI [22] integrates visual-tactile fingertips, a wrist force/torque sensor, and pose tracking in a handheld gripper, and uses the collected modalities for contact-event segmentation. PrintTac targets the remaining connection between portable collection and repeatable deployment. Its magnetometer measurements are sent by a Wi-Fi microcontroller to the same smartphone that records RGB observations and ARKit poses, producing a self-contained demonstration stream. More importantly, our cross-instance experiment keeps the learned policy fixed when the deployed tactile fingertips are replaced. This test links manufacturing consistency to the embodied behavior learned from the portable interface.

III. MULTIMODAL SENSING WITH PRINTTAC

A. Hardware Design

The PrintTac tactile fingertip is engineered around a tripartite architecture: an onboard sensing and wireless transmission core, a deterministic compliant flexure that maps contact wrenches to displacement, and a mechanically decoupled, application-specific contact interface (Fig. 1).

Compliant Flexure and Material Selection: To bypass the manufacturing variability of cast elastomers, PrintTac produces deterministic 6-DoF motion at the contact surface through a monolithic 3D-printed flexure inspired by planar compliant mechanisms [23], while the resulting surface motion is implicitly encoded in magnetic-field variations measured by the magnetometer array. Each finger integrates two symmetric, semi-circular compliant springs rigidly anchored to the gripper chassis at their centers, with their outer perimeters driving the contact interface through triple wave-shaped flexure beams. This dual-spring topology enhances out-of-plane torsional rigidity and mitigates irreversible plastic deformation compared to cantilevered single-spring baselines.

Material selection is critical to ensuring elastic repeatability. Thermoplastic polyurethane (TPU) exhibits severe non-linear hysteresis and collapses under modest loads, drastically curtailing dynamic range. Conversely, standard polylactic acid (PLA) offers high initial stiffness but suffers from brittle failure and persistent viscoelastic creep under cyclic

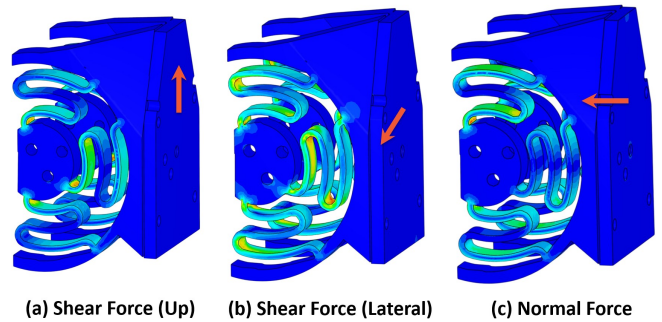


Fig. 2: **Finite element analysis of the PrintTac flexure.** Strain distributions under a 30 N load applied in three directions: (a) upward shear, (b) lateral shear, and (c) normal compression.

loads. We adopt polyethylene terephthalate glycol-modified (PETG), which combines high fatigue yield strength with rapid elastic recovery, thereby bounding viscoelastic drift. Static finite element analysis (FEA) under rated multiaxial loading (Fig. 2) confirms that stress is distributed relatively uniformly across the wave-shaped beams without localized peak concentrations, suggesting that the structure remains within its linear elastic regime under the tested loading conditions.

Sensing Core and Decoupled Modular Fingertips:

Beneath the flexure, there is a board containing an array of five three-axis magnetometers. We use a magnetometer board used in eFlesh [10]. Each magnetometer uses the Hall effect to measure the magnetic field along three orthogonal axes (b^x, b^y, b^z). The magnetometer board is interfaced to an Adafruit QT Py ESP32 microcontroller. Cylindrical neodymium permanent magnets are seated within recessed cavities on the rear surface of the rigid contact plate. Deflections of the compliant frame shift these magnets relative to the static magnetometer array, altering the localized magnetic flux density.

Crucially, the external contact fingertip is fabricated from rigid PLA and mechanically fastened via threaded fasteners, completely decoupling the contact interface from the internal magnetic sensing stack. This modularity allows operators to swap wear surfaces or adapt contact geometries in under one minute without disassembling the sensing core or recalibrating the magnetic baseline. We leverage this versatility across distinct tasks, pairing high-friction flat elastomers for dynamic wiping with sharp, rigid geometries for micro-object grasping (Fig. 3).

IV. UNTETHERED MULTIMODAL DEMONSTRATION SYSTEM

A. System Architecture and Wireless Edge Integration

To support high-throughput, in-the-wild data collection, we build an untethered handheld demonstration suite integrated into the Universal Manipulation Interface (UMI) paradigm [11]. Existing tactile teleoperation setups typically tether operators to stationary host PCs via high-bandwidth USB or DAQ cables, severely restricting human ergonomics.

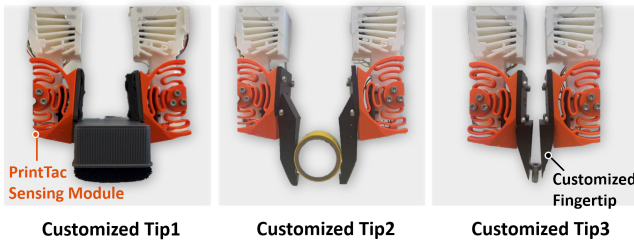


Fig. 3: **Modular fingertip adaptations.** Diverse contact geometries tailored for specific manipulation primitives (e.g., planar wiping pads vs. pointed micro-pinching tips), all interchangeably mounted on the identical PrintTac sensing base.

In our design, all visual, tactile, and kinematic streams are ingested, synchronized, and logged directly onboard a handheld commercial smartphone (Apple iPhone).

The physical interface mounts two PrintTac fingers onto a 3D-printed parallel gripper chassis. Each sensing finger is powered directly via an inline USB-C pass-through adapter connected to the iPhone battery, drawing negligible current. Sensor measurements sampled by the onboard ESP32 are packetized and broadcast over a local ad-hoc Wi-Fi hotspot hosted by the phone, decoupling the operator from any external network infrastructure.

B. Temporal Alignment and Kinematic Calibration

Data integrity during demonstration recording relies on strict multimodal synchronization and spatial hand-eye calibration:

Observation Synchronization: The custom iOS application concurrently processes three asynchronous observation streams:

- *Spatial Trajectories:* Real-time 6-DoF end-effector poses derived from the Apple ARKit VIO/SLAM pipeline at 60 Hz.
- *Visual Streams:* High-resolution RGB video captured by the primary camera at 60 Hz. A secondary ultra-wide camera simultaneously tracks stereo ArUco fiducials mounted on the finger links to infer continuous gripper aperture via a calibrated linear projection.
- *Tactile Feedback:* The 200 Hz incoming UDP magnetic flux stream is monotonically timestamped against the system clock and downsampled via linear interpolation to lock-step alignment with the 60 Hz camera shutter.

Video and timestamped sensor data are stored locally on the smartphone, then uploaded to a host computer for offline alignment and conversion into Zarr training datasets.

Hand-Eye Calibration: To accurately project demonstration trajectories onto the robot’s physical coordinate frame, we perform offline kinematic hand-eye calibration. The smartphone is mounted rigidly to the robot tool flange, which executes an automated calibration trajectory through non-planar SE(3) configurations. By aligning paired ARKit poses with ground-truth robot forward kinematics, we resolve the fixed rigid-body transformation ${}^E\mathbf{T}_C$. The calibrated

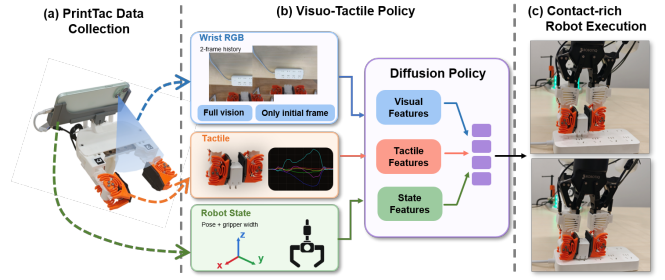


Fig. 4: **Pipeline of the multimodal conditional diffusion policy.** Visual frames from the wrist camera are encoded via a ResNet-18 backbone, while normalized low-dimensional tactile vectors and proprioceptive states are concatenated directly without a dedicated encoder. The unified multimodal embedding conditions a 1D convolutional U-Net to denoise action trajectories in a receding-horizon fashion.

transformation achieves an average translation residual under 2 mm and rotational drift under 0.5° , confirming that ARKit trajectories provide sufficient tracking fidelity for fine-grained contact tasks.

V. VISUO-TACTILE POLICY WITH PRINTTAC UMI

To leverage the multi-axis contact signals from PrintTac, we formulate visuo-tactile manipulation using conditional Diffusion Policy [1]. Instead of treating tactile feedback as an auxiliary high-dimensional visual stream, we use the raw magnetic field measurements as a compact tactile observation, without a dedicated encoder or pose-reconstruction module.

A. Multimodal Policy Architecture

The policy architecture adopts a conditional 1D convolutional U-Net that denoises an action sequence conditioned on an observation history window (Fig. 4). At each control timestep t , the multimodal observation comprises three components:

- *Visual Observations:* An observation history $\mathbf{O}_t = \{o_{t-1}, o_t\}$ comprising the two most recent RGB frames captured by the wrist-mounted camera. Each frame is processed by a ResNet-18 visual backbone [24] to extract a compact visual embedding.
- *Tactile Observations:* A temporal window of multi-axis magnetic flux measurements $\mathbf{T}_t = \{t_{t-1}, t_t\}$, where each t_τ concatenates the 3-axis readings across the magnetometer channels of the tactile finger. Crucially, unlike vision-based tactile approaches that require complex pre-trained encoders (e.g., tactile ViTs or CNNs), $\mathbf{T}_t$ is normalized with training-set statistics and concatenated directly into the state vector, without a dedicated tactile encoder.
- *Proprioceptive State:* The robot’s current proprioceptive state $\mathbf{P}_t$, including continuous gripper aperture.

The flattened visual embeddings, normalized tactile readings, and proprioceptive features are concatenated into a flattened global conditioning vector $\mathbf{c}_t = [\text{Enc}(o_{t-1}, o_t), \mathbf{T}_t, \mathbf{P}_t]$,

which modulates the intermediate features of the 1D U-Net via Feature-wise Linear Modulation [25].

B. Action Representation and Closed-Loop Execution

Following the UMI formulation [11], the policy predicts a sequence of n future end-effector waypoints $\mathbf{A}_t = \{a_{t+1}, \dots, a_{t+n}\}$. Each action waypoint $a_\tau = (\Delta \mathbf{T}_\tau, \mathbf{w}_\tau)$ consists of a 6-DoF end-effector pose relative to the current tool frame along with the absolute gripper opening width $\mathbf{w}_\tau \in [0, 1]$. Formulating trajectories in the local relative coordinate frame improves cross-environment spatial generalization.

The policy is executed in a receding-horizon control loop. Although the diffusion model generates an entire chunk of n future steps, the robot executes only the immediate k steps ($k \leq n$) before re-sampling observations and re-conditioning the policy. This continuous closed-loop replenishment allows the policy to react dynamically to instantaneous contact events—such as jamming during tight pin-slot insertion or slippage during grasping—at high control frequencies.

Observation Modality Variants: In addition to the standard closed-loop setting where $\mathbf{O}_t = \{o_{t-1}, o_t\}$ updates continuously, we investigate a *vision-deprived tactile control* setting in the tactile only wiping task in Section VI-B. Under this scenario, visual input is restricted strictly to the initial scene frame ($\mathbf{O}_t = \{o_0\}$), forcing the diffusion policy to guide fine-grained action entirely through PrintTac tactile dynamics.

VI. EXPERIMENTS

Our empirical evaluation investigates the physical sensing fidelity of PrintTac and its real-world utility for contact-rich robotic manipulation by addressing four core research questions:

- Q1: Force-Estimation Fidelity.** *How accurately can the magnetic measurements predict normal and shear forces under held-out loading trajectories?*
- Q2: Fabrication consistency.** *Does additive manufacturing reproducibility preserve the magnetic-to-force mapping across independently fabricated units, enabling zero-shot calibration reuse without per-instance retraining?*
- Q3: Contact-Phase Identifiability.** *Can low-dimensional PrintTac observations unambiguously distinguish critical contact transitions throughout dynamic manipulation sequences?*
- Q4: Closed-Loop Policy Utility.** *Does incorporating tactile feedback improve task success rates over vision-only policies?*

A. Force Estimation and Fabrication Consistency

To answer **Q1** and **Q2**, we benchmark both within-instance force-estimation accuracy and zero-shot cross-instance calibration transfer. We independently 3D-print and assemble two distinct PrintTac elastomers (S_A and S_B) following the identical fabrication procedure. A multiaxial load cell records ground-truth forces over dynamic loading protocols ranging from 0 to 25 N across both normal and shear

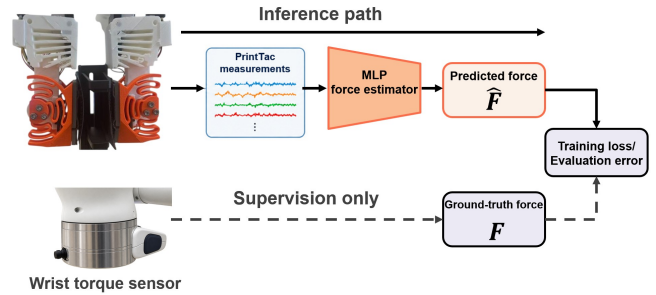


Fig. 5: **Model architecture of force-estimation task.** An MLP maps PrintTac magnetic measurements to the predicted contact force. Ground-truth measurements from the wrist force–torque sensor are used only for training supervision and evaluation, and are not required during inference.

directions (Fig. 5). A lightweight multi-layer perceptron (MLP) is trained strictly on measurement sequences from instance S_A . We first quantify within-instance accuracy on held-out loading trajectories collected on S_A (**Q1**), and subsequently deploy this unmodified model directly onto the unseen instance S_B without per-device recalibration, fine-tuning, or baseline offset adjustment (**Q2**).

Normal-Force Estimation: Under held-out within-instance testing on S_A , the model achieves a Mean Absolute Error (MAE) of 0.345 N, corresponding to a Normalized MAE (NMAE) of 1.38% over the full 25 N operational dynamic range (Fig. 6(a) Top). When evaluating this frozen model on the independently fabricated instance S_B , the MAE shifts to 0.705 N (NMAE: 2.82%, Fig. 6(a) Bottom). Crucially, despite the complete absence of sensor recalibration, the cross-instance MAE increases by 0.360 N, equivalent to 1.44% of the 25 N force range. This minor drift confirms that our monolithic 3D-printed flexure successfully maintains predictable force-displacement kinematics across separate fabrication cycles.

Shear-Force Estimation: Under shear loading, the estimator yields an MAE of 0.196 N (NMAE: 0.78%) on held-out trials from S_A (Fig. 6 (b) Top). When evaluated directly on the unseen instance S_B , the model retains exceptional fidelity, yielding an MAE of 0.202 N (NMAE: 0.81%, Fig. 6 (b) Bottom). The cross-instance error increase is negligible (0.006 N, or 0.024% of the dynamic range), demonstrating that the lateral dual-spring geometry enforces tight mechanical constraints against parasitic transverse displacements.

Taken together, these evaluations confirm that PrintTac provides high-fidelity force feedback for manipulation (**Q1**) and that its physical response profile is preserved across independently fabricated units, eliminating the per-device calibration bottleneck in tactile learning workflows (**Q2**).

B. Contact-Rich Manipulation

To address **Q3** and **Q4**, we analyze the physical contact signatures and multimodal trajectory dynamics across three representative contact-rich manipulation tasks: plug insertion, delicate chip picking, and compliant whiteboard wiping (Fig. 7). These benchmarks encompass critical manipulation

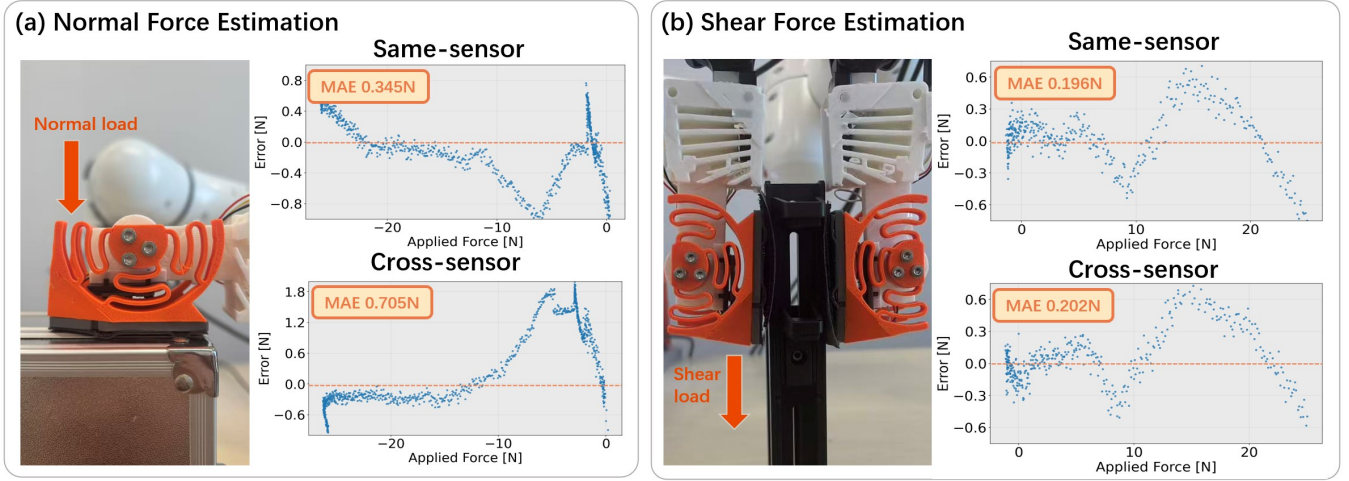


Fig. 6: **Experimental setup for force-estimation task.** The images illustrate our experimental setup, in which normal forces are generated by pressing and tangential forces are generated through friction. We map the raw sensor readings to force magnitudes using an MLP and compare the predicted forces with the ground-truth forces. For each type of force, we conduct two evaluation settings, with the results shown in the figure. The upper plots show the prediction error when the same sensor is used for both training and testing, but under different force application trials. The lower plots show the prediction error when training and testing are performed on independently fabricated sensing elastomers.

primitives, including initial impact detection, fragile force regulation, sustained surface compliance, and geometric constraint alignment. For each benchmark, we temporally align the multichannel magnetic trajectories with the robot kinematics, demonstrating how PrintTac yields unambiguous, phase-dependent observations that inform closed-loop policy execution.

1) *Plug Insertion:* The robot must insert a pre-grasped electrical plug into a tight-tolerance power socket. The execution requires initial downward descent, tangential planar search along the socket face plate, and dynamic downward force regulation to overcome internal spring-loaded shutter resistance without mechanical jamming.

Contact Signature and Policy Behavior: We collect 65 demonstrations with small random spatial perturbations applied to the socket location to enforce spatial generalization. As illustrated in Fig. 7(a), the synchronized magnetic profile distinctly isolates three sequential phases:

- 1) *Surface Impact:* An immediate positive normal deflection occurs when the plug lands on the power strip surface.
- 2) *Alignment Search:* As the end-effector glides along the plastic casing, steady shear-force signals register friction until the prongs mate with the socket receptacle.
- 3) *Snap-Through Insertion:* The moment of alignment is marked by a sharp reduction in vertical reaction force as the prongs drop into the socket cavities, followed by a steep increase in downward normal force as the policy drives the plug through internal mechanical latches.

In contrast to the visuo-tactile policy, the vision-only baseline suffers from severe occlusions caused by the gripper fingers and plug housing. Lacking force feedback to detect prong-

socket alignment, the vision-only agent sometimes jams against the casing and remains stalled.

2) *Chip Picking:* The manipulator approaches and grasps ultra-thin, brittle potato chips from a flat substrate. The policy must apply sufficient gripping force to prevent the chip from slipping without exceeding the chip’s fracture limit, followed by an immediate upward retrieval.

Contact Signature and Policy Behavior: We collect 63 demonstrations spanning chips of diverse form factors, surface finishes, and support pedestal elevations. As shown in Fig. 7(b), the transition from free-space finger closure to physical contact produces a distinct, high-gradient impulse in the magnetic readings. Conditioned on this contact onset, the visuo-tactile policy immediately arrests lateral finger closure and transitions to the vertical lift phase, continuously modulating gripper aperture to hold grasping force within a calibrated non-destructive margin. Policies lacking tactile sensing cannot detect zero-clearance surface contacts, leading to either dropped objects from premature stopping or shattered components from excessive motor drive.

3) *Compliant Whiteboard Wiping:* The robot manipulates an eraser across a whiteboard surface to clear random dry-erase markings. Crucially, the policy must regulate a consistent normal contact force across varying surface inclinations. Insufficient normal force leaves residual ink, whereas excessive contact force triggers the manipulator’s built-in joint torque safety thresholds.

Contact Signature and Policy Behavior: We collect 68 demonstrations on horizontal boards and evaluate on surfaces tilted up to 15° . Because surface inclination and depth discrepancies are heavily distorted under monocular wrist-camera perspectives, visual feedback alone is insufficient to maintain uniform surface contact. As depicted in the temporal traces in Fig. 7(c), the initial contact impact triggers an

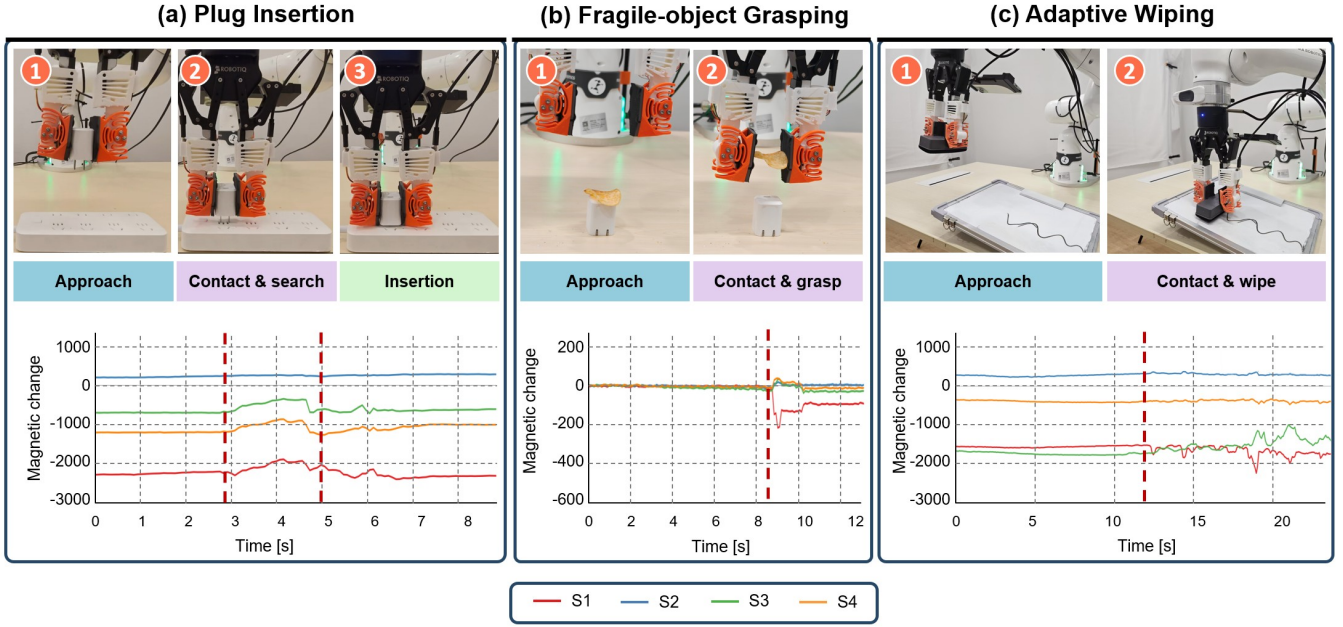


Fig. 7: **Qualitative rollout trajectories and contact-phase signatures across contact-rich manipulation tasks.** Benchmarks are evaluated on a Franka Research 3 manipulator fitted with a Robotiq 2F-85 gripper and PrintTac fingertips. In each rollout, temporal magnetic trajectories (displaying four prominent magnetometer channels for visual clarity) clearly demarcate distinct contact transitions: (a) impact, surface search, and snap-through insertion during plug mating; (b) soft contact landing and sustained pinch regulation during delicate chip picking; and (c) constant normal preload and stick-slip friction oscillations during tilted surface wiping. Note that only 4 magnetometer readings are shown for concision, and each reading is the sum of the three measured field components.

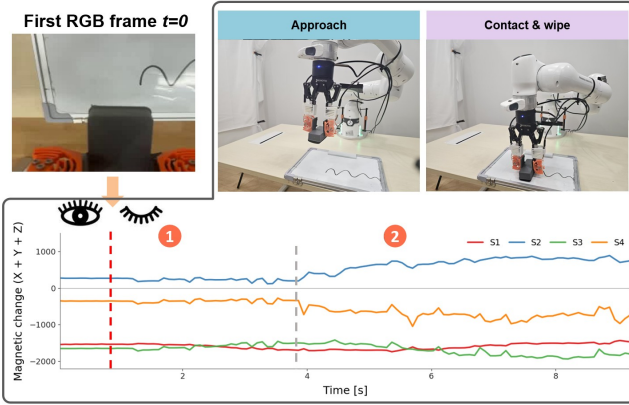


Fig. 8: **Vision-deprived surface wiping evaluation.** The policy receives only the initial static RGB image ($t = 0$) to estimate rough workspace bounds; the entire subsequent descent, surface contact detection, and planar wiping sequence are governed exclusively by closed-loop PrintTac feedback.

abrupt step-change in magnetic flux, followed by steady-state oscillations reflecting dynamic stick-slip friction as the felt eraser traverses the board. Conditioned on this continuous feedback, the policy dynamically adjusts its Cartesian vertical trajectory to track the inclined contour while maintaining consistent normal pre-load.

4) *Wiping without Online Visual Feedback:* To evaluate whether low-dimensional PrintTac observations can support reactive manipulation when visual perception is completely

severed, we conduct a vision-deprived experiment on the wiping task (Fig. 8).

We collect 39 demonstrations with variable board heights. During closed-loop rollout, the policy is conditioned strictly on the initial visual snapshot $\mathbf{O}_t = \{o_0\}$ captured while the arm is hovering in free space. The control loop must therefore execute the approach, contact detection, and planar cleaning entirely through closed-loop proprioception and tactile feedback. As verified in the trajectories, the policy maintains an open-loop downward plunge until PrintTac detects surface collision via an abrupt step-change in magnetic flux. Upon registering contact, the policy immediately halts vertical descent and smoothly transitions into tangential wiping passes, actively regulating the normal force across the unknown surface height. This demonstrates that PrintTac provides physically sufficient, unambiguous contact cues capable of driving reactive behaviors even under extreme perceptual degradation.

C. Contact-Rich Manipulation Policy Performance

We compare policies trained with and without tactile observations using the same demonstration datasets, policy backbone, and number of training epochs. Both policies receive the same visual and proprioceptive inputs; the visuotactile policy additionally receives PrintTac measurements.

For each task, we evaluate each policy over 10 trials under matched scene configurations and robot initial poses. A plug-insertion trial succeeds when more than 80% of the plug’s insertion length enters the socket. Chip picking succeeds

Task	Vision Only	Vision + PrintTac
Plug Insertion	40% (4/10)	90% (9/10)
Chip Picking	20% (2/10)	100% (10/10)
Whiteboard Wiping	30% (3/10)	90% (9/10)
Tactile-Only Wiping	0% (0/10)	90% (9/10)

TABLE I: **Task success with and without tactile input.** Policies with PrintTac input achieve higher success rates than policies without tactile input in the evaluated tasks. Note that policy trained on demonstrations collected with S_{demo} transfer directly to S_{eval} without per-device recalibration.

when the robot grasps the chip and lifts it to the designated location. Whiteboard wiping succeeds when more than 90% of the visible markings are removed. Trials that do not meet the success criterion within 45 seconds are counted as failures.

We also evaluate wiping without online visual feedback. In this setting, both policies receive only the initial image throughout the episode, while proprioceptive observations continue to update. The visuo-tactile policy also receives updated tactile measurements. The success criterion is the same as for whiteboard wiping.

As shown in Tab. I, adding tactile input improves success rates on all four tasks. Success increases from 40% to 90% for plug insertion, from 20% to 100% for chip picking, and from 30% to 90% for whiteboard wiping. Without online visual feedback, the policy with tactile input succeeds in 9 of 10 trials, whereas the policy without tactile input fails in all 10 trials. These results support the benefit of tactile feedback for the evaluated contact-rich tasks, addressing Q4.

Demonstrations are collected using a UMI gripper equipped with sensor instances S_{demo} , while robot evaluation uses separately fabricated instances S_{eval} . The policies are deployed without retraining. Successful execution with S_{eval} provides evidence of policy transfer across sensor instances under the tested conditions, addressing Q2.

VII. CONCLUSION AND FUTURE WORK

We present PrintTac, a tactile fingertip that combines a 3D-printed compliant structure, off-the-shelf sensing components, and a replaceable contact tip. We integrate it into a portable demonstration-collection interface and evaluate its use in robot learning. Force-estimation models transfer across separately fabricated fingertips, and policies using tactile observations achieve higher task success rates than policies without tactile input.

Long-term durability and drift under sustained or repeated loading remain to be characterized. Future work will also address magnetic interference and reduce the size of the compliant structure. By simplifying fabrication and supporting the reuse of tactile data and learned models across sensor instances, PrintTac lowers practical barriers to scaling tactile robot learning.

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